VIOLATIONLEDGER: GPS-BASED GEOLOCATION PARKING INFRACTION DETECTION WITH SMART ANALYTICS AND ESCALATION FOR BARANGAY BLUE RIDGE B

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In Partial Fulfillment
of the Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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APPROVAL SHEET

In partial fulfillment of the requirements for the degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY,** this Capstone project entitled **"VIOLATIONLEDGER: GPS-BASED GEOLOCATION PARKING INFRACTION DETECTION WITH SMART ANALYTICS AND ESCALATION FOR BARANGAY BLUE RIDGE B"**, has been prepared and submitted by **Alboladora, Emily A., Ampusta, Jerald F., Bitong, Nicole V., Cinco, Adrian I., Dioso, Jane Claudeth P., Guisijan, Rizaleda Gabriela C., Macaranas, Ameerah T., Morva, Nathaniel Carlo M., Pajermo, Jason M., Robante, James D., Trero, Melanie Mae M., Zuela, John Lenard C.,** who are hereby recommended for project presentation.

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DEDICATION

ACKNOWLEDGEMENT

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(---------PROJECT BRIEF------------------------------)
SUMMARY OF CHAPTER 1 TO 5

TABLE OF CONTENTS

Title Page i

Approval Sheet....................................... ii

Dedication iii

Acknowledgement iv

Executive Summary....................................... v

Table of Contents....................................... vi

List of Tables....................................... viii

List of Figures....................................... ix

List of Appendices....................................... x

Chapter 1 Capstone Project Background

Introduction

Project Context and Its Background

Purpose and Description

Objective of the Study

Main Objective

Specific Objectives

Scope and Limitation/Delimitations of the Project

Theoretical Framework and Conceptual Framework of the Study

Definition of Terms

Chapter 2 Review of Related Literature, Studies and Systems

Related Literature

Local Literature

Foreign Literature

Related Studies

Local Literature
Foreign Literature
Technical Background
Chapter 3 Design and Methodology
Methodology
Requirement Analysis
Sampling Technique
Current Technical Situation
Requirement Documentation
Project In-scope
Project Out-scope
Design of Software, Systems, Product and/or Processes
Development and Testing
Implementation Plan

References

Appendices ....

List of Tables

Table No.	Table Title	Page No.
1.1	Hardware Components	33
1.2	Software Components	35
1.3	Implementation Plan	48

List of Figures

Figure No.	Figure Title	Page No.
1.1	Theoretical Framework	17
1.2	Conceptual Framework	20

List of Appendices

Appendix	Appendix Title	Page No.
A	Curriculum Vitae	55

CHAPTER I

CAPSTONE PROJECT BACKGROUND

Introduction

Parking violations are a common problem in many barangays, particularly in Barangay Blue Ridge B where parking spaces are limited, resulting in improper parking. This often causes traffic congestion, blocked passageways, and safety risks for pedestrians and drivers. Traditional approaches, such as manual patrolling and issuing tickets, require significant time and effort from barangay officials and are prone to errors and inconsistencies. With the continuous increase of vehicles in local communities, a faster and more reliable system is needed to address parking violations.

This study introduces ViolationLedger: a GPS-based system for detecting parking infractions with smart analytics and escalation. The system uses GPS technology to track vehicles and determine whether they are parked in no-parking zones or restricted areas. Once a violation is detected, the system records the incident, sends an SMS notification to the driver, and escalates the case if no action is taken within a specified time.

The system also maintains a database of vehicles and their registered owners, allowing barangay officials to properly identify violators. In addition, the system generates reports showing common violations, streets with the highest number of cases, and whether violations are done repeatedly. These insights can help barangay leaders create better policies and improve compliance among residents and visitors.

Overall, ViolationLedger offers a practical and modern approach to parking enforcement at the barangay level. It reduces the burden on manpower, improves accuracy, and promotes fairness, while ensuring safer and more organized streets for the community.

Project Context and Its Background

Barangay Blue Ridge B is a residential area in Quezon City under District III, with its current Barangay Captain, Esperanza Castro Lee, as the chief administrator of the community. The barangay is responsible for maintaining order and safety within its jurisdiction, which includes the regulation of traffic rules, parking, and road use within the barangay premises. At present, violation monitoring and parking regulation in the barangay are largely handled manually by enforcers and barangay personnel. This manual process is prone to inefficiency, delays, and human error, which often lead to disputes and challenges in documenting and enforcing violations.

One of the recurring issues faced by the barangay is the inefficient detection and documentation of parking infractions. The current method relies on human observation and physical recording, which is not only time-consuming but also subject to inconsistencies and lack of transparency. Residents and vehicle owners may contest the validity of certain violations due to the absence of systematic evidence. Furthermore, the lack of an integrated record-keeping system makes it difficult for the barangay to analyze recurring traffic and parking problems or escalate unresolved issues effectively.

The goal of the proposed system, ViolationLedger: GPS-Based Geolocation Parking Infraction Detection with Smart Analytics and Escalation, is to address these concerns by introducing a technology-driven approach to violation monitoring. By leveraging GPS-based geolocation, the system can automatically detect and record parking infractions within the barangay premises with minimal human intervention. The integration of smart analytics will allow for accurate documentation of violations, generation of violation trends, and systematic escalation for unresolved or repeat offenses.

The proposed solution may significantly benefit Barangay Blue Ridge B by providing a reliable and transparent platform for violation detection and monitoring. With automated tracking and smart analytics, barangay officials can make data-driven decisions, reduce the administrative burden on enforcers, and ensure fair and consistent enforcement of parking rules. For the residents, the system introduces accountability, faster resolution of disputes, and an assurance that violations are properly recorded and managed with supporting digital evidence.

By implementing ViolationLedger, the barangay can modernize its approach to community governance, improve efficiency in handling parking violations, and strengthen the trust of residents in the enforcement system.

Project Purpose and Description

ViolationLedger aims to provide a modern and organized solution to the problem of illegal parking in the barangay. Such violations often lead to traffic congestion, blocked passageways, and safety risks for both pedestrians and drivers. The traditional forms of handling these violations, that is, manual patrolling and ticketing, take so much time, effort, and labor from barangay officials and do not translate well in regard to errors and inconsistencies.

The project was developed to assist the barangay officials in solving such cases in the shortest time possible. The system can automatically detect vehicles parked in restricted areas, through a GPS-based tracking mechanism, record the violations, and notify the owner. If there is no response from the driver, the violation may be referred for proper resolution. The system also keeps a database of vehicles and residents that enable easy identification of violators.

It contains reports about the most common violations, streets where such violations happen, and the specific persons who repeatedly violate the rules. This helps to form better rules within the community and to foster fairness and discipline within the community. Although ViolationLedger would never replace the authority and judgment of barangay officials, it stands as a reliable support system that lightens the burden, increases accuracy, and contributes to uniform enforcement of rules. Most importantly, it brings safety and orderliness to the streets while giving benefits both to the community residents and visitors.

Objective of the Study

Main Objective

The ViolationLedger project aims to provide consistent, timely, and organized solutions to identify and manage illegal parking violations in barangay by utilizing GPS-based tracking mechanisms, intelligent analytics and escalation. The system aims to provide a modern, efficient, and advanced solution to help avoid illegal parking, reduce traffic congestion, and promote safer organized streets.

Specific Objectives

The proposed system sought to achieve the following objectives:

1. To develop and integrate a GPS-based tracking module to monitor and identify vehicles parked in restricted areas.

2. To create a violation detection mechanism that automatically records and timestamps infractions, including vehicle location and duration of the violation.

3. To create and establish a centralized database for storing vehicle and owner information, allowing for easy identification of repeat offenders.

4. To develop an automated notification and escalation that sends an initial SMS warning to the owner of the vehicle and escalates the case if the vehicle owner is not responsive.

Scope and Delimitations of the Project

Scope

The system covers the following features and boundaries:

1. The system assists barangay officials in managing parking violations by reducing manual effort, improving accuracy, and ensuring fairness in enforcement.

2. The system utilizes GPS-based detection to identify vehicles parked in no-parking zones or restricted areas, automatically recording location, time, and duration of violations.

3. The system provides an SMS notification feature to inform vehicle owners of their infractions, with an escalation process for unresolved cases.

4. The system maintains a centralized database that stores vehicle and owner information, enabling barangay officials to identify violators and track repeat offenses effectively.

5. The system offers analytics and reporting tools to highlight common violations, identify hotspot areas, and track frequent offenders, while also providing predictive analytics to anticipate high-risk times and locations.

6. The system enforces security through role-based access control, granting access exclusively to authorized barangay officials and administrators.

Limitations/Delimitations

The system does not cover features outside the stated scope. Specifically:

1. The system is designed exclusively for barangay-level use and does not extend to citywide or nationwide traffic enforcement.

2. The system is limited to GPS-based detection and does not integrate third-party hardware such as CCTV cameras or speed detectors.

3. The system does not cover payment processing of fines or citation fees, as these remain under the jurisdiction of existing barangay procedures.

4. The system is developed only for iOS and Android mobile platforms and does not include a web-based or desktop version.

5. The effectiveness of the system depends on the accuracy of registered vehicle and owner information, the stability of mobile networks for SMS notifications, and the reliability of GPS signals in specific locations.

6. The system functions as a support tool to aid barangay officials in enforcing regulations but does not replace their authority or legal responsibility in handling violations.

Theoretical Framework and Conceptual Framework of the Study

Theoretical Framework

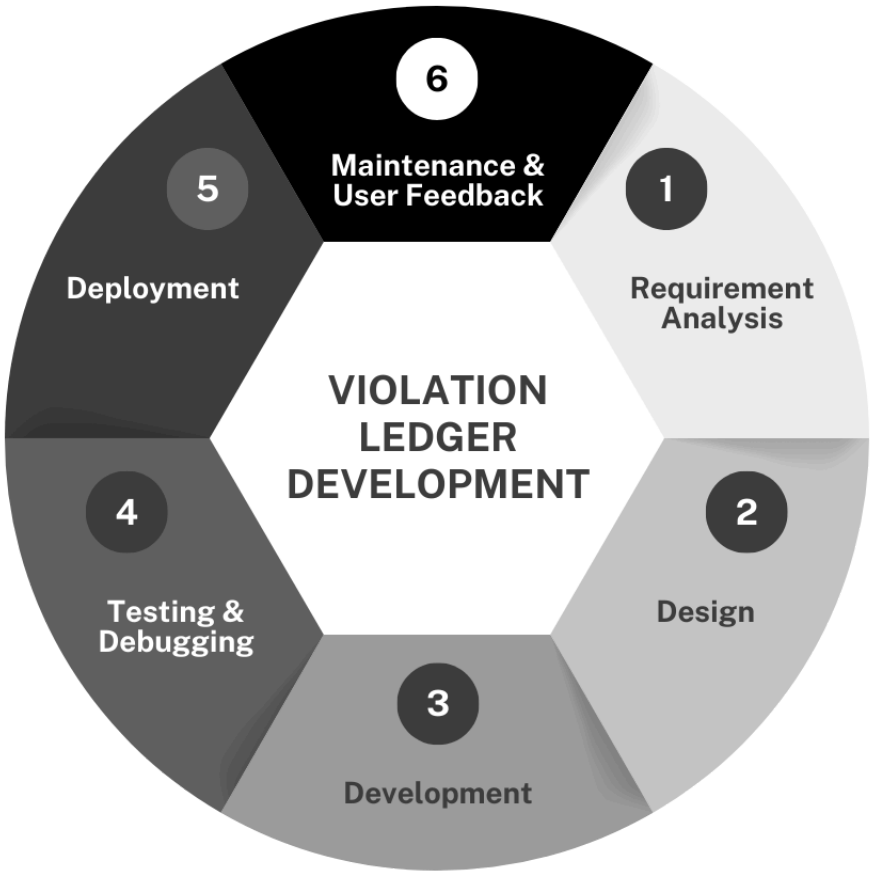

Figure 1.1 *Agile Framework for ViolationLedger: GPS-Based Geolocation Parking Infraction Detection With Smart Analytics and Escalation*

The Agile Framework was adopted for the development of ViolationLedger because it emphasizes flexibility, collaboration, and continuous improvement. Unlike rigid project management methods, Agile allows the system to be developed in iterative cycles, ensuring that every stage of progress is refined based on feedback from barangay officials and users. This approach ensures that the system remains responsive to real-world needs while minimizing risks and addressing challenges early in the process. Through Agile, ViolationLedger can adapt to changing requirements and incorporate enhancements over time, making it an ideal framework for building a community-focused enforcement system.

The first phase is Requirement Analysis, where the needs of the barangay and the challenges in parking enforcement are identified. Guided by deterrence theory, this stage emphasizes the necessity of certainty, swiftness, and severity in the design of the system. The requirements gathered focus on real-time GPS monitoring, automated SMS notifications, and escalation processes to effectively address parking violations.

The second phase is Design, where the system requirements are translated into a structured model. At this stage, the system’s architecture is created with features that align with deterrence principles, such as GPS-enabled tracking, violation logging, and escalation mechanisms. The Agile framework plays a key role by ensuring that the design remains flexible and adaptable to adjustments based on feedback from stakeholders.

The third phase is Development, where the system is built iteratively. Developers create and integrate modules for geolocation, SMS alerts, and escalation workflows. Each cycle of development reflects deterrence in practice, ensuring certainty through monitoring, swiftness through immediate notifications, and severity through structured escalation. The Agile approach enables continuous progress and refinement with every iteration.

The fourth phase is Testing and Debugging, where the system undergoes thorough testing to detect and correct errors. This step guarantees that GPS monitoring is accurate, SMS alerts are promptly delivered, and escalation processes function as intended. The Agile framework reinforces continuous improvement by making testing a recurring activity after each cycle, thereby ensuring the system's reliability and effectiveness in reinforcing deterrence.

The fifth phase is Deployment, where the ViolationLedger system is introduced into the barangay setting. During this phase, deterrence principles are fully applied in real-world conditions: GPS monitoring begins detecting violations, SMS notifications are delivered in real time, and escalation mechanisms activate when violations remain unresolved. This phase validates the system's capability to translate theoretical foundations into practical enforcement.

The final phase is Maintenance and User Feedback, which ensures the system's sustainability and adaptability over time. This phase reflects the Dynamic Theory of Deterrence and Compliance (Bekar & Carlaw, 2022), as consistent monitoring, visible enforcement, and regular updates reinforce long-term compliance. Feedback from barangay officials and residents is continuously integrated through the Agile framework, allowing the system to evolve and remain effective in addressing illegal parking.

Conceptual Framework

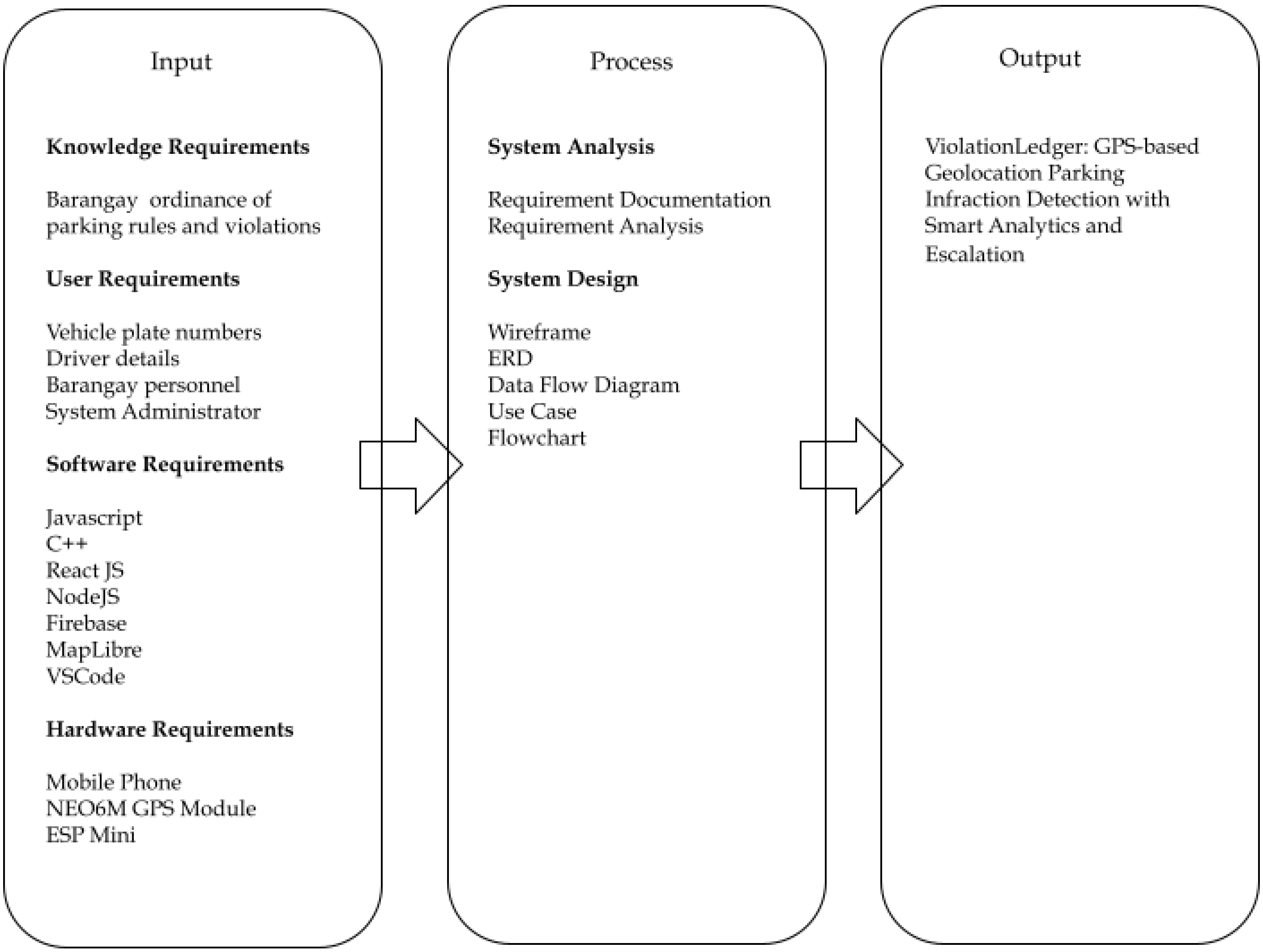

Figure 1.2 *IPO Model for ViolationLedger: GPS-Based Geolocation Parking Infraction Detection With Smart Analytics and Escalation*

Input

Knowledge Requirements

The proponents studied the Barangay ordinance of parking rules and violations to understand the legal framework and specific guidelines on prohibited parking behaviors. Research and consultations with barangay officials were conducted to identify how violations are currently monitored, recorded, and addressed, as well as the challenges encountered with manual enforcement. This knowledge provided a foundation for designing a system that automates detection, ensures compliance with regulations, and minimizes reliance on human intervention.

User Requirements

The system requires inputs such as vehicle plate numbers and driver details for proper identification of violators. It also depends on barangay personnel for validation and monitoring, alongside a system administrator who manages the records, analytics, and escalation processes. These requirements ensure that the system is practical and accessible for both users and enforcement authorities.

Software Requirements

The proponents utilized several programming languages and frameworks to develop the system. JavaScript, C++, ReactJS, and NodeJS were used for the application logic and interface. Firebase was selected for real-time database management and cloud storage, while MapLibre was integrated for geolocation mapping and visualization. The development process was supported using Visual Studio Code (VSCode) as the primary Integrated Development Environment (IDE). These tools established the foundation for a responsive, scalable, and data-driven solution.

Hardware Requirements

To support system functionality, the proponents employed mobile phones for end-user access, a NEO6M GPS module for real-time geolocation tracking, and an ESP Mini for lightweight processing and integration. These hardware components enable accurate detection of parking infractions, seamless communication with the system, and reliable operation within the barangay setting.

Process

System Analysis

The proponents performed a detailed requirement analysis to ensure the system effectively addressed the challenges of manual parking violation monitoring. Documentation of system needs was carried out to identify potential issues such as delayed reporting, inconsistent enforcement, and lack of transparency. This analysis guided the design of a system that minimizes human error, enhances accuracy, and streamlines the monitoring of parking infractions.

System Design

The system design phase involved the creation of wireframes, entity-relationship diagrams (ERD), data flow diagrams, use case models, and flowcharts. These design tools established a structured blueprint for developers and stakeholders, ensuring that the system's features aligned with the project objectives while optimizing data flow, geolocation processing, and interface usability for efficiency and real-world application. GPS, however, faces limitations such as reduced accuracy in areas with poor satellite visibility, signal interference, or delays in position updates. To address these, the ViolationLedger system integrates smart analytics to validate location data, verifies repeated violations to minimize false detection, and applies escalation mechanisms that maintain accurate and reliable enforcement despite these constraints.

System Development

Following the design stage, the proponents implemented core functionalities through iterative development cycles. Developers created prototypes, integrated the GPS module, and refined the system through continuous testing. Wireframes guided the user interface design, while ERDs and flowcharts directed backend integration. Testing was performed at each stage to verify reliability, scalability, and usability. Iterative improvements were applied consistently, ensuring that the final product addressed user feedback and community requirements.

Output

The expected output of this study is the ViolationLedger: GPS-based Geolocation Parking Infraction Detection with Smart Analytics and Escalation. This system automates the detection of parking violations using GPS technology, provides real-time SMS alerts to violators, and applies escalation procedures for unresolved infractions. By combining smart analytics with automated monitoring, the system enhances transparency, reduces the workload on barangay personnel, and promotes discipline among residents. Ultimately, the system aims to improve order, accountability, and compliance within Barangay Blue Ridge B.

Definition of Terms

AI-based Traffic Violation Detection (Related Concept) - Systems employing artificial intelligence to identify traffic violations, reducing human labor and increasing enforcement accuracy. While relevant to the field, this concept is broader than the ViolationLedger's scope.

Automatic Number Plate Recognition (ANPR) - An AI-based technology that identifies vehicles by reading license plates for enhanced monitoring and enforcement accuracy.

Automated Logging - The process by which detected parking infractions are automatically recorded in a system database without manual intervention.

Blockchain Technology (Related Concept) - A secure and transparent digital ledger system that ensures tamper-proof logging and fair allocation of resources. While significant, it is considered a related technology and not the primary foundation of ViolationLedger.

Crowd-Sensing (Related Concept) - A method using aggregated GPS data from multiple users or devices to detect patterns of illegal parking. Though relevant, it is classified as a related concept rather than a direct component of ViolationLedger.

Current Technical Situation - Understanding how the organization currently conducts its business processes to effectively design software solutions that fit these real-world contexts.

Database - A structured digital storage system containing vehicle information and owner details used for accurate identification of violators.

Deterrence Theory - A criminological theory stating that people are less likely to commit violations if punishment is certain, swift, and sufficiently severe, serving as the foundation for the ViolationLedger system.

Dynamic Theory of Deterrence and Compliance - An updated perspective explaining that ongoing, visible enforcement over time increases compliance by reinforcing the perceived risk of detection and punishment.

Escalation Process - The procedure by which unresolved parking violations are forwarded for further action or stricter enforcement.

GPS Tracking - Technology that uses satellite signals to determine and monitor the precise location of vehicles to identify if they are parked in restricted areas.

Illegal Parking - The act of parking a vehicle in areas designated as no-parking zones or restricted spaces, which often leads to traffic congestion, blocked passageways, and safety concerns.

Manual Patrolling - Traditional method of managing parking violations involving physical monitoring and ticketing by barangay officials.

Predictive Analytic - The use of data analysis techniques to forecast parking violation patterns, helping barangay officials anticipate high-risk times and areas.

Real-Time Monitoring - Continuous observation and data collection on parking and traffic activities to enable immediate response to violations.

Requirement Documentation - A detailed enumeration of software features describing what the system will do, answering "how the software will perform" without detailing design elements like database structure or architecture.

Sampling Technique - The strategy used to select a subset (sample) from the entire population under study, which influences the generalizability of the findings.

Smart Parking Systems - Integrated technological solutions employing sensors, networks, and data management to optimize parking enforcement and reduce urban congestion.

S**MS Reminder -** A text message sent automatically by the ViolationLedger system to notify a driver about a detected parking violation.

Violation Report - A generated summary by the system showing patterns of parking violations,such as the most common offenses, frequently offending vehicles, and hotspot locations for illegal parking.

ViolationLedger - A GPS-based digital system designed to detect parking infractions automatically, record violations, notify drivers via SMS, and escalate cases if unresolved, aimed at improving parking enforcement at the barangay level.

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

The system combines GPS tracking, automated logging, SMS notification, and predictive analytics to improve parking enforcement. A database stores vehicle and owner information for accurate identification of violators. To support its design, related literature and studies were reviewed. Literature provides theories on traffic management and technology use, while studies show practical applications using GPS, AI, and mobile systems. These serve as a guide in developing ViolationLedger.

FOREIGN LITERATURE

Smart Parking Systems

Biyik et al. (2021) emphasize the growing importance of smart parking systems in addressing urban congestion. They outline a layered architecture consisting of sensors, networks, data management, and applications, showing how these integrate for efficient enforcement. The authors note that while sensor- and camera-based systems provide accuracy, GPS and crowd-sensing methods are more cost-effective in resource-limited contexts. Their work supports ViolationLedger reliance on GPS-based detection and layered design, while reinforcing the value of analytics for predicting violations.

Number Plate Recognition

Obaid et al. (2022) examine the role of Automatic Number Plate Recognition (ANPR) in modern smart city systems. They highlight advancements in artificial intelligence and cloud-based solutions that improve recognition speed and accuracy, enabling wider use in traffic enforcement and parking management. The review underscores how ANPR contributes to real-time monitoring and compliance tracking. This literature suggests that ViolationLedger could expand beyond GPS detection to include automated identification for more precise enforcement.

Crowd-Sensing for Parking Patrol

He et al. (2022) demonstrate the potential of crowd-sourced GPS trajectories as a tool for monitoring parking compliance. They emphasize how aggregated mobility data from bike-sharing systems can reveal patterns of illegal parking and guide enforcement more efficiently. This literature highlights the feasibility of using geolocation data as a scalable enforcement tool, directly aligning with the ViolationLedger goal of GPS-based infraction detection.

AI and Blockchain in Parking

Khan et al. (2024) propose PARKTag, which integrates artificial intelligence with blockchain for trusted parking management. Their work underscores the importance of secure and transparent systems that reduce fraud and enhance accountability. By showing how blockchain ensures tamper-proof records and fair allocation of spaces, this literature provides insights for ViolationLedger in considering secure data handling and transparent escalation processes.

Blockchain for Space Allocation

Zhang et al. (2024) highlight how blockchain technology can improve the fairness and efficiency of private parking allocation. They demonstrate that smart contracts and decentralized consensus can prevent disputes and maximize space utilization. For ViolationLedger, this literature suggests that blockchain can strengthen credibility by ensuring violation records remain secure, transparent, and beyond manipulation.

LOCAL LITERATURE

AI for Violation Apprehension

Bayangos et al. (2021) introduce the application of artificial intelligence in automating traffic violation detection in the Philippines. Their work shows the benefits of reducing human intervention while improving consistency in enforcement. This literature supports ViolationLedger by showing that AI-driven enforcement is already being applied locally, and that automation is both feasible and effective.

Computer Vision for Parking

Garcia et al. (2024) highlight the use of computer vision to monitor and manage parking spaces in real time. Their study shows how automation can minimize disputes and improve order in shared parking areas. This literature illustrates how vision-based monitoring can complement geolocation systems, suggesting a possible hybrid approach for ViolationLedger in the future.

Residents' Parking Preferences

Villanueva et al. (2018) explore residents' preferences regarding parking facilities in Davao City, finding that accessibility, affordability, and security significantly shape acceptance. This literature emphasizes the importance of public perception in the success of parking systems. For ViolationLedger, it highlights that enforcement solutions must also consider user trust and accessibility to gain community support.

Violert System

Tagaloguin et al. (2022) introduce Violert, a system using image recognition to detect parking violations in paid facilities. Their work emphasizes how automation can reduce reliance on manual inspection and improve compliance. This literature strengthens ViolationLedger direction by showing that local initiatives already apply intelligent detection, making its proposed automation both practical and relevant.

Occupancy and License Plate Monitoring

Galanao et al. (2020) develop a monitoring system that combines occupancy detection with license plate identification. Their approach enables real-time tracking of vehicles and improved enforcement of parking rules. This literature demonstrates how combining geolocation or occupancy monitoring with vehicle identification enhances accountability, providing ViolationLedger with a model for future system upgrades.

FOREIGN STUDIES

AI-Driven Parking Enforcement System

According to Tan and Lee (2021), an AI-driven parking enforcement system in Singapore using CCTV reduced manual errors in violation detection. This is related to ViolationLedger because both promote automation, though ViolationLedger uses GPS instead of CCTV, making it more practical for barangays without large-scale camera networks.

Predictive Analytics for Smart Parking

According to Shin and Park (2021), predictive analytics for smart parking in South Korea allowed forecasting of peak violation times. This supports

ViolationLedger because its predictive analytics feature also helps barangays anticipate and prepare for high-risk periods.

Real-Time Parking Violation Detection Using Deep Learning

According to Peng, Zhao, and Sun (2022), a real-time parking violation detection system using deep learning automated the identification of parked vehicles. This is connected to ViolationLedger because both aim to minimize human effort and improve efficiency, even though one uses cameras and the other uses GPS.

Crowd-Sensing with GPS Trajectory Data

According to He, Yan, and Wang (2021), crowd-sensing methods using GPS trajectory data were effective in detecting illegal parking. This strongly supports ViolationLedger because both rely on geolocation to detect violations and manage enforcement.

Blockchain-Based Parking Enforcement System

According to Ahmed, Khan, and Al-Hammadi (2022), a blockchain-based parking enforcement system in the UAE ensured transparency through automatic logging and secure records. This relates to ViolationLedger because both systems emphasize accountability, even though ViolationLedger does not apply blockchain technology.

LOCAL STUDIES

AI-Based Contactless Traffic Violation Apprehension System

According to Jose et al. (2021), an artificial intelligence-based contactless traffic violation apprehension system was developed in the Philippines to reduce human effort in monitoring traffic rules. This is related to ViolationLedger because both automate violation detection, although ViolationLedger uses GPS technology instead of AI cameras.

Barangay Profiling and Certification System

According to Ballaran, Duran, and Villanueva (2023), a barangay profiling and certification system was created using regression analysis to improve the management of resident records. This is connected to ViolationLedger since both utilize digital databases to increase efficiency and accountability, with ViolationLedger focusing on vehicle and violator records.

Multi-Level Car Parking Facility Feasibility Study

According to Co, Polinar, and Payao (2022), a feasibility study on a multi-level car parking facility in Davao City revealed that the shortage of parking spaces contributes to illegal parking. This supports ViolationLedger because it addresses violations even when infrastructure improvements, such as new parking facilities, are not yet available.

Vehicle Parking Watch System

According to Trinidad and Porquis (2024), the “Vehicle Parking Watch” system in Iligan City effectively monitored parking activities and improved compliance. This relates to ViolationLedger because both are localized systems that enhance enforcement at the barangay level.

Mobile Applications for Traffic Enforcement

According to Delos Reyes and Ramos (2021), mobile applications for traffic enforcement in Metro Manila reduced delays and provided faster access to records. This aligns with ViolationLedger because it also uses a mobile-based GPS platform with automated escalation to make violation management more proactive.

Technical Background

Hardware Components

The ESP32 Mini and NEO-6M GPS module are the core hardware components of the GPS-Based Geolocation Parking Infraction Detection System, working together to provide real-time vehicle tracking with accurate GPS data, wireless communication, and efficient power usage, making them ideal for detecting parking violations.

Hardware	Specification	Descriptions
Microcontroller (ESP32 Mini)	ESP32-WROOM-32, Low Power Mode, Wi-Fi & Bluetooth Connectivity	A small, low-power microcontroller with integrated Bluetooth and Wi-Fi is the ESP32 Mini. It will communicate with the mobile app or backend (using REST API or Sockets.io) to transmit GPS data and violations.

GPS Module (NEO-6M)	NEO-6M GPS Module, Ublox Chipset, Accuracy of ~2.5m	The vehicle location will be tracked in real time using the NEO-6M GPS module. It provides a good accuracy level for geolocation and is commonly used in projects.
Power Supply/Battery Pack	Li-Ion/Li-Po battery (3.7V–5V) - Rechargeable with TP4056 charging module	Stable power is provided to the ESP32 and peripherals for fixed deployments.

Table 1.1 *Hardware specifications required for the implementation of the ViolationLedger system.*

Software Components

The mobile app is developed using React Native, which offers a smooth and responsive user interface for both iOS and Android. The backend is powered by Node.js, which also manages communication between the server and the mobile application and real-time data processing. Firebase handles cloud functions and user authentication in addition to storing and synchronizing data in real-time. In order to visualize and identify parking infractions, MapLibre is integrated for interactive mapping, which shows the locations of vehicles and geofenced parking zones. These software elements combine to form a system that is effective, scalable, and easy to use.

Software	Specification	Descriptions
Mobile App: React Native	Frontend Framework: GPS Tracking, Geofencing, Firebase Integration, Push Notifications	React Native is used in the cross-platform development of the mobile application (iOS and Android). It tracks GPS locations, detects parking violations, and integrates with Firebase to send push notifications and sync data in real time
Backend: Node.js	Backend Framework: REST API, Socket.io (Real-Time Communication), Firebase Integration	The backend is handled by Node.js. It handles violation reports, uses Socket.io to communicate with the mobile app in real-time, and integrates with Firebase for database management and user authentication.

Database: Firebase	Database: Firebase Firestore, Firebase Authentication, Firebase Cloud Messaging (FCM)	All of the app data, including GPS coordinates, violation histories, and user information, is stored and managed using Firebase. Firebase Authentication handles user authentication, while FCM notifies officers via push notifications.

Table 1.2. *Software specifications required for the implementation of the ViolationLedger system.*

CHAPTER III
DESIGN AND METHODOLOGY

This chapter discusses the Design and Methodology of the hardware and software requirements that were used in developing and implementing the project. It would also discuss the required analysis and documentations in order to develop the system.

Methodology

The research methodology employed in this study is developmental research, with the primary objective of designing, developing, and refining an innovative system that addresses real-world challenges in parking violation detection and enforcement. The focus is on improving the effectiveness and efficiency of monitoring in Barangay Blue Ridge B through the development of the proposed system, ViolationLedger: GPS-based Geolocation Parking Infraction Detection with Smart Analytics and Escalation. This study takes advantage of modern technological tools, particularly geolocation and smart analytics, and investigates their impact on both enforcement efficiency and the satisfaction of residents and barangay personnel.

Developmental research is characterized by its emphasis on the systematic design, testing, and improvement of a solution. The process follows a structured sequence: identifying the challenges faced by the barangay in enforcing parking regulations, designing system components tailored to these needs, developing the prototype, and testing the system in realistic scenarios. This staged approach ensures that outcomes are not only innovative but also continuously refined based on actual performance and user feedback.

A mixed-method approach will be used by combining both quantitative and qualitative techniques. Quantitative data will be collected through surveys distributed to residents and barangay personnel to measure system usability, timeliness of violation alerts, and satisfaction levels. Metrics such as accuracy of detection, response times, and frequency of successful escalations will be objectively measured using a confusion matrix, system log timestamps, and violation database records. At the same time, qualitative data will be obtained through interviews and open-ended feedback, allowing participants to share their experiences and perspectives regarding the effectiveness and ease of use of the system. This feedback will highlight areas for improvement that may not be captured by quantitative measures.

The collected data will undergo analysis to compare system outcomes with current manual enforcement methods. This comparison will determine whether ViolationLedger provides significant improvements in usability, efficiency, and user satisfaction. Emphasis will be placed on identifying benefits such as reduced reliance on manual monitoring, faster alert dissemination, and improved compliance among residents.

Throughout the research process, continuous refinement will be carried out based on empirical evidence and user insights. This ensures that the system evolves into a reliable and accessible solution. Ultimately, this developmental research bridges theory and practice by producing an operational system that directly addresses the challenges of parking violation enforcement in Barangay Blue Ridge B. The findings are expected to contribute not only to improved local governance but also to future research and the advancement of technology-driven community enforcement systems.

Requirement Analysis

The following were identified during the requirement analysis stage regarding the current system of Barangay Blue Ridge B:

WHO - The people involved (stakeholders & roles)

- Barangay Captain & Council/Barangay Officials - policy owners and approvers; decision makers.
- Barangay Secretary/Administrator - maintains records, coordinates escalations and reports.
- Residents/Vehicle owners - affected users; recipients of SMS warnings and subjects of analytics.
- Homeowners' Association (HOA) Officers - consulted during approvals and community rollout.
- Proponents/Project team/IT support - system builders, trainers, and implementers.

WHAT - The business activities and information handled today

- Manual monitoring and patrolling of streets and parking areas; issuing verbal warnings or physical tickets when found.
- Physical recording of violations (not centralized), causing lack of evidence.
- Incident escalation when violators do not comply (handled by barangay officials following local ordinance).

WHERE - The environment/context where work happens

- Barangay Blue Ridge B, a residential subdivision in Quezon City (District III). Narrow roads / limited parking. The system is scoped to barangay-level deployment (not city-wide).

WHEN - Timing, schedules, frequency and temporal constraints

- Current enforcement is ad-hoc based on patrol schedules and reported complaints (manual). The proposed system will provide real-time detection with pilot runs and phased rollout: approvals (2–3 days), community consultation (1 day), system prep (1–2 days), pilot test (3–5 days), training (2–3 days), full deployment (1 day).

HOW - How current procedures are actually performed

- Observation by enforcers (patrols) → manual recording (paper/ledger) → escalation done manually if the owner is not responsive. There is no automated GPS / geofence evidence collection currently; this causes disputes and inconsistent records.

WHY - The organization uses a traditional method for violation checking

- Low infrastructure budget for fixed CCTV or expensive sensors - manual patrolling is the low-capex default.
- Legal & community practices - enforcement via barangay tanods and barangay procedures has established social acceptance and known escalation paths; changing processes needs community buy-in.
- Data & trust concerns - records have historically been local, human-readable logs; full digital change requires trust-building.

- Human oversight is valued - the system is explicitly designed to support barangay officials, not replace their judgment, so manual control remains culturally and legally relevant.

Sampling Technique

For the requirement analysis, the proponents used a purposive sampling technique. This method was chosen because only specific individuals directly involved in the current parking violation monitoring system of Barangay Blue Ridge B can provide accurate and relevant information. The respondents included barangay officials such as the Barangay Captain, council members, the secretary, and assigned tanods, since they are responsible for enforcement and record-keeping. In addition, selected residents and vehicle owners were consulted to capture their experiences and perspectives regarding the effectiveness and limitations of the current system.

This targeted approach ensured that the data collected reflected the actual processes, challenges, and needs of the organization, thereby forming a reliable basis for designing the proposed system.

Requirement Documentation

The system ViolationLedger-GPS Context-Aware Geolocation Parking Infraction Detection with Predictive Analytics and Automated Escalation, is intended to automate parking violation detection, provide real-time notifications, generate violation analytics, and support predictive enforcement.

In – Scope

Login and Logout Module - This module allows users to enter their username and password to access the system.

Parking Violation Detection Module - This module detects illegal parking using GPS coordinates in restricted or no-parking zones.

Notification Module - This module sends SMS alerts to the violators once a parking violation is detected.

Analytics Module - This module provides reports and dashboards showing violation trends creating heatmaps of areas with many violations and uses data to predict when and where violations may occur.

User Roles and Access Control Module - This module is responsible for managing access to the system by ensuring that only authorized users are able to utilize specific features.

Out - Scope

Integration with third-party enforcement hardware - The system will not connect directly to external devices such as CCTV cameras, speed guns, or other enforcement tools.

Payment processing for fines or citation fees - Handling or processing of payments related to parking fines or violations will not be managed within the system.

Web-based platform development - A browser-based (web) version of the system will not be developed as part of this project.

Designs of Systems, Software, Products, and/or Processes

This section focuses on the technical design reflecting how requirement analysis and documentation have been addressed. It explains any tradeoffs considered due to constraints and may include conceptual designs, data models, system architecture diagrams, or block diagrams demonstrating the ability to identify, formulate, and solve computing problems.

Development and Testing

During the implementation phase, the design concepts of ViolationLedger were translated into executable codes. The system was developed using React Native for the mobile application, Node.js for the backend, and Firebase for real-time database management and cloud functions. MapLibre API was integrated to provide geolocation and mapping visualization, while the ESP32 Mini microcontroller and NEO-6M GPS module were configured to supply accurate location data for detecting parking infractions. The choice of these technologies was made to align with the specific requirements of the application in the development of a GPS-Based Geolocation Parking Infraction Detection System with Smart Analytics and Escalation. The primary goals included enhancing efficiency, reducing manual enforcement, and ensuring security and reliability in handling sensitive vehicle and owner information.

In the evaluation and testing phase, the system underwent rigorous verification to ensure that all modules aligned with user-defined requirements. Testing was carried out following ISO 25010 quality standards, focusing on functionality, usability, reliability, and security.

Functional Suitability. The system was designed to automatically detect vehicles parked in restricted zones using GPS data, record violations with timestamps, and send SMS notifications to owners. It also manages escalation when warnings are ignored and provides analytics on violation trends. Each feature was tested to ensure accuracy and ease of use for barangay officials and enforcers.

Performance Efficiency. The system processes GPS data in real time and generates notifications within seconds. During pilot testing, the app reliably logged violations, displayed heatmaps, and produced reports without noticeable delays. This ensures that officials can respond quickly to parking infractions while minimizing system load and processing errors.

Compatibility. ViolationLedger was tested across multiple devices, ensuring compatibility with both Android and iOS platforms. The system integrates seamlessly with MapLibre for geofencing and visualization and works on mobile devices commonly used by barangay personnel. Its design ensures accessibility regardless of device specifications or operating systems.

Usability. The interface of the mobile application was created to be simple and intuitive, allowing barangay enforcers to log in, view violations, and access reports with minimal training. Buttons and menus were designed with clarity, ensuring smooth navigation for non-technical users. Pilot participants rated usability highly on the Likert scale, confirming that the system is user-friendly.

Reliability. The system demonstrated consistent performance during extended operation, handling multiple simultaneous GPS inputs without downtime or crashes. Accurate and up-to-date violation data was maintained in the Firebase database, reducing the risk of lost or inconsistent records. Reliability testing also confirmed the system's stability in areas with variable mobile signal strength.

Security. Role-based access control was implemented to restrict system features to authorized users only. Firebase Authentication and Cloud Messaging ensured that sensitive data—such as vehicle and owner information—was securely stored and transmitted. Encryption protocols were tested to safeguard against unauthorized access, reinforcing community trust in the system.

Maintainability. The system's modular architecture, built with React Native and Node.js, allows for easy updates and improvements. This structure ensures that future features, such as blockchain-based violation records or AI-enhanced detection, can be integrated with minimal rework. Maintenance processes such as error logging, system monitoring, and database updates were also tested to guarantee smooth long-term functionality.

Portability. ViolationLedger can be accessed from various mobile devices, enabling flexibility and convenience for enforcers and officials. Whether on smartphones or tablets, the platform provides real-time access to violation data, notifications, and analytics. This portability ensures that the system can be used effectively in different barangay environments and potentially expanded to other communities.

Data Analysis Plan

To evaluate the effectiveness, usability, and reliability of the proposed system, the proponents will employ both descriptive and inferential statistical tools. The survey questionnaires are designed using a five-point Likert scale, where responses are quantified as follows:

- 5 – Strongly Agree / Very Good / Very Willing / Very High

- 4 – Agree / Good / Willing / High
- 3 – Undecided / Fair
- 2 – Disagree / Hesitant / Low / Poor
- 1 – Strongly Disagree / Very Hesitant / Very Low / Very Poor

The following statistical methods will be applied:

1. Weighted Arithmetic Mean
   a. Used to determine the average perception of respondents on each survey item.
2. Standard Deviation
   a. Measures the variability of responses around the mean, showing whether perceptions are consistent or widely spread.
   b. A lower standard deviation indicates strong agreement among respondents, while a higher value suggests diverse opinions.
3. Analysis of Variance (ANOVA)
   a. Applied to determine if there are significant differences in responses among different groups of respondents (e.g., barangay officials, enforcers, and residents).
   b. This helps evaluate whether perceptions vary depending on the role of the respondent.
4. Pearson Product-Moment Correlation
   a. Used to assess the relationship between variables, such as the correlation between usability and user satisfaction or between system performance and efficiency of violation handling.

b. The value of the correlation coefficient (ranging from -1 to +1) will indicate the strength and direction of the relationship.

Implementation Plan

The implementation of the ViolationLedger: GPS-Based Parking Infraction Detection System in Barangay Blue Ridge B will be carried out in phases to ensure both technical readiness and community acceptance. It begins with securing approval from barangay officials and homeowners' association leaders, followed by resident consultations to build transparency and address concerns. After approval, the system will be prepared and tested through pilot runs in selected streets to verify GPS accuracy, violation logging, and SMS notifications.

Once validated, the project will proceed with an information campaign and training sessions for barangay personnel to ensure proper use and monitoring of the system. Full deployment will then be implemented across the subdivision, with continuous evaluation to refine processes, identify violation hotspots, and strengthen compliance. This structured approach ensures the system not only enforces parking rules effectively but also promotes order and safety within the community.

Strategy	Activities	Person Involved	Duration
Pre-Implementio n Approval	Conduct initial meetings with the Barangay Captain, council, and HOA representatives; Submit formal letters of intent and project proposal	Proponents, Barangay Officials, HOA Officers	2–3 days
Community Consultation	Present system overview to residents during barangay assembly or HOA meeting; Gather feedback and address concerns	Proponents, Barangay Officials, Residents	1 day
System Preparation	Installation of server, configuration of mobile application, and	Proponents, IT Support	1–2 days

	setup of GPS modules; Ensure system is compatible with subdivision existing resources		
Pilot Testing	Limited rollout in selected streets/zones to test GPS accuracy, SMS notifications, and escalation process	Proponents, Barangay Officials, Assigned Enforcers	3–5 days
Information Campaign	Distribution of flyers and posters in subdivision bulletin boards, gates, and common areas; Orientation sessions with residents to explain rules and system use	Barangay Officials, HOA, Residents	2–3 days

Training	Training barangay personnel on app use, violation logs, and report generation.	Proponents, Barangay Enforcers, Barangay Secretary	2–3 days
Full Deployment	Activate system for the entire subdivision; Monitor real-time reports and SMS notifications	Proponents, Barangay Officials	1 day
Monitoring & Evaluation	Regular review of violation trends, system performance, and resident feedback; Adjust escalation process as needed	Barangay Council, HOA, Proponents	Ongoing (monthly/ quarterly)

Table 1.3. *Implementation strategies and activities for the deployment of the ViolationLedger system.*

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APPENDICES

A. Curriculum Vitae

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PERSONAL INFORMATION

Date of Birth: August 18, 2003

Place of Birth: Quezon City

Citizenship: Filipino

Gender: Female

Civil Status: Single

Interest: Physical fitness, music, food, crafting

EDUCATIONAL BACKGROUND

Primary: Pasong Tamo Elementary School

Secondary: Culiat High School

Emilio Jacinto High School

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Determined
- Adaptability
- Resilient
- Team-Oriented

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PERSONAL INFORMATION

Date of Birth: October 4, 2002

Place of Birth: Quezon City

Citizenship: Filipino

Gender: Male

Civil Status: Single

Interest: Technology and Innovation, Blockchain Gaming & Web3, Video Editing and Graphic Design, Community Events & Public Engagement, Programming

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EDUCATIONAL BACKGROUND

Primary: Manuel L. Quezon Elementary School

Secondary: Commonwealth High School – Junior High School

Commonwealth High School – Senior High School (HUMSS Strand)

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Organized
- Efficient
- Interpersonal skills
- Communication skills
- Computer proficient
- Video editing

QUEZON CITY UNIVERSITY

NICOLE V. BITONG

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PERSONAL INFORMATION

Date of Birth: June 4, 2004

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Citizenship: Filipino

Gender: Female

Civil Status: Single

Interest: Music, Adventures, Reading,Music

EDUCATIONAL BACKGROUND

Primary: Pres.Corazon C. Aquino Elementary

Secondary: Batasan Hills National High School

Batasan Hills Senior High School

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Computer Skills
- Communication
- Time Management
- Good Attitude

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PERSONAL INFORMATION

Date of Birth: December 16, 2001

Place of Birth: Southern Leyte

Citizenship: Filipino

Gender: Male

Civil Status: Single

Interest: Watching anime, Gaming, Sports

EDUCATIONAL BACKGROUND

Primary: Rosa L. Susano Novaliches Elementary

Secondary: San Bartolome High School

Bestlink College of the Philippines

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Teamwork
- Collaboration
- Strong Attention to Detail
- Flexible

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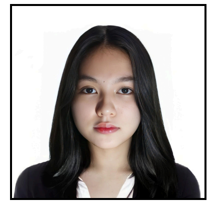

PERSONAL INFORMATION

Date of Birth: June 21, 2004

Place of Birth: Lupi, Camarines Sur

Citizenship: Filipino

Gender: Female

Civil Status: Single

Interest: Movies, Singing, Mobile Games

EDUCATIONAL BACKGROUND

Primary: Sauyo Elementary School

Secondary: Sauyo High School

Gardner College

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Basic understanding of computer support and troubleshooting technical issues.
- Strong leadership and organizational skills, with the ability to manage teams and complete projects on time.
- Effective logical and professional problem-solving abilities.
- Experienced in project management, UI/UX design, and quality analysis, with a strong emphasis on achieving high-quality outcomes.

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EDUCATIONAL BACKGROUND

Primary: Sauyo Elementary School

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Bestlink College of the Phillipines

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PERSONAL QUALIFICATION

- Technical Skills
- Basic Mathematical
- Customer Service
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Gender: Female

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Interest: Reading, Writing, Arts

EDUCATIONAL BACKGROUND

Primary: Pasong Tamo Elementary School

Secondary: Sauyo High School

St. Augustine College of Nursing Quezon City

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Strong collaboration and teamwork skills
- Effective communicator, both visual and verbal
- Creative and adaptable in problem-solving
- Detail-oriented with a passion for design and the arts

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Gender: Male

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Interest: Coding, Gaming, Reading, Travelling

EDUCATIONAL BACKGROUND

Primary: GSIS Village Elementary School

Secondary: Ismael Mathay Sr. High School

Villagers Montessori College

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Full Stack Developer
- Adaptable
- Flexible
- Willing to learn

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Gender: Female

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Interests: Adventure, Music, Games, Sports

EDUCATIONAL BACKGROUND

Primary: San Gabriel Elementary School

Secondary: Sta.Lucia High School

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PERSONAL QUALIFICATION

- Computer Skills
- Troubleshooting
- Good Attitude

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Date of Birth: March 13, 2004

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Gender: Male

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Interest: Reading manga, Playing Guitar, Games.

EDUCATIONAL BACKGROUND

Primary: Commonwealth Elementary School

Secondary: North Fairview High School

South East- Asia Institute of Trade and Technology

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Computer Skills
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- Good Attitude

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Date of Birth: March 6, 2003

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Gender: Female

Civil Status: Single

Interest: Dancing, Gaming, Reading, Sports

EDUCATIONAL BACKGROUND

Primary: San Gabriel Elementary School

Secondary: Holy Spirit National High School

STI College Fairview

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Adaptable
- Eager to learn
- Open to feedback
- Positive Attitude

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Date of Birth: July 9, 2002

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Gender: Male

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Interest: Reading, Music, Games

EDUCATIONAL BACKGROUND

Primary: Angelique Divine Grace School

Heuin Dol World Mission Christian Academy

Secondary: San Bartolome High School

Metro Manila College

Tertiary: Quezon City University (Current)

PERSONAL QUALIFICATION

- Creative
- Open To Criticism
- Digitally Knowledgable